Práctica Tema 3
Servidor de Aplicaciones

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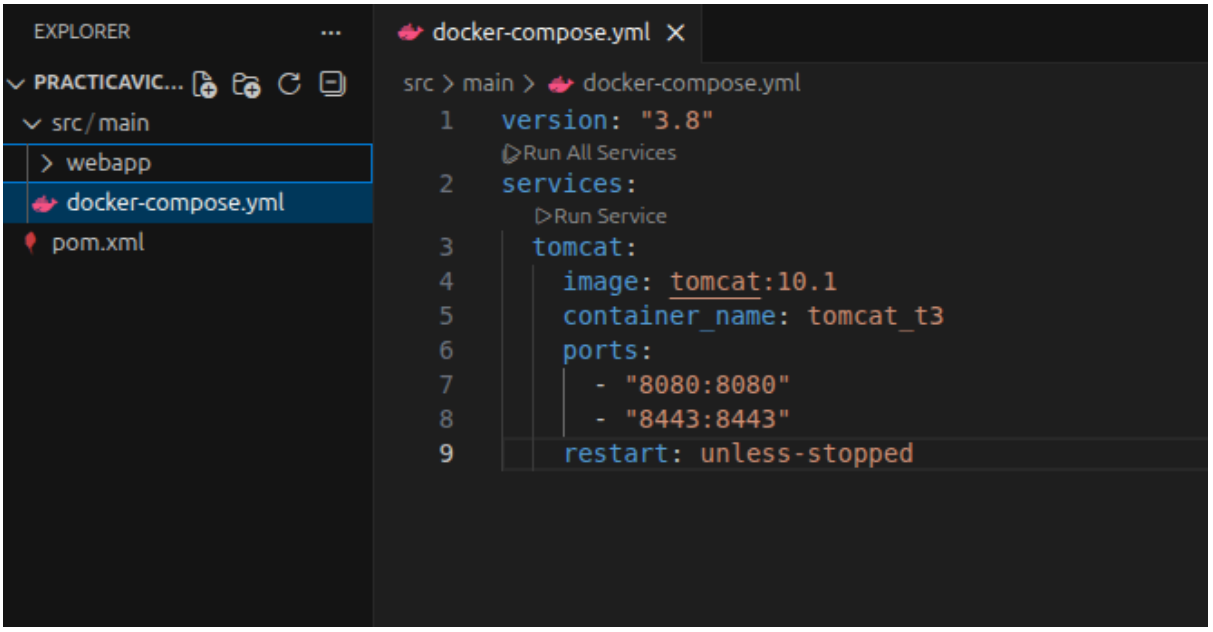
Paso 1



Estado inicial del proyecto y de la receta de docker

Lo levanto pero me da error:

```
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor$ sudo docker compose up -d
[sudo] password for victrespy:
no configuration file provided: not found
```

El error es porque no estoy en la carpeta del archivo docker y me tengo que mover a ella

```
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main$ sudo docker compose up
    ✔ 8be760c2a9a8 Pull complete
    ✔ 4f4fb700ef54 Pull complete
    ✔ d0702afe109d Pull complete
    ✔ ae2fca99c299 Pull complete
    ✔ eabd93b5d461 Pull complete
[+] Running 2/2
 ✔ Network main_default  Created
 ✔ Container tomcat_t3   Started
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main$ sudo docker ps
CONTAINER ID   IMAGE         COMMAND             CREATED         STATUS         PORTS
                                                   NAMES
397060ab7c92   tomcat:10.1   "catalina.sh run"   8 seconds ago   Up 7 seconds   0.0.0.0:8080->8080/tcp, [::]
->8080/tcp, 0.0.0.0:8443->8443/tcp, [::]:8443->8443/tcp   tomcat_t3
```

Paso 2

Instalo los certificados

```
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main/certs$ openssl req -x509 -no
des -days 365 -newkey rsa:2048 -keyout tomcat.key -out tomcat.crt
.+.+........+.......+..+.+...........+...+......+.+......+............+...+..+......+......++++++++++++++++++++++
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:
State or Province Name (full name) [Some-State]:
Locality Name (eg, city) []:
Organization Name (eg, company) [Internet Widgits Pty Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:
Email Address []:
```

Genero los archivos y actualizo la receta de docker

```
EXPLORER
PRACTICAVICTOR
  src/main
    certs
      tomcat.crt
      tomcat.key
    webapp
  docker-compose.yml
  pom.xml

docker-compose.yml
src > main > docker-compose.yml
1  version: "3.8"
   Run All Services
2  services:
     Run Service
3    tomcat:
4      image: tomcat:10.1
5      container_name: tomcat_t3
6      ports:
7        - "8080:8080"
8        - "8443:8443"
9      volumes:
10       - ./certs:/usr/local/tomcat/certs
11     restart: unless-stopped
```

Uso el siguiente comando para comprobar que funciona

```
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main$ sudo docker exec -it tomcat
_t3 ls /usr/local/tomcat/certs
tomcat.crt  tomcat.key
```

Paso 3

```
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main$ sudo apt install maven
Adding debian:Telia_Root_CA_v2.pem
Adding debian:TeliaSonera_Root_CA_v1.pem
Adding debian:TrustAsia_Global_Root_CA_G3.pem
Adding debian:TrustAsia_Global_Root_CA_G4.pem
Adding debian:Trustwave_Global_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P256_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P384_Certification_Authority.pem
Adding debian:T-TeleSec_GlobalRoot_Class_2.pem
Adding debian:T-TeleSec_GlobalRoot_Class_3.pem
Adding debian:TUBITAK_Kamu_SM_SSL_Kok_Sertifikasi_-_Surum_1.pem
Adding debian:TunTrust_Root_CA.pem
Adding debian:TWCA_Global_Root_CA.pem
Adding debian:TWCA_Root_Certification_Authority.pem
Adding debian:UCA_Extended_Validation_Root.pem
Adding debian:UCA_Global_G2_Root.pem
Adding debian:USERTrust_ECC_Certification_Authority.pem
Adding debian:USERTrust_RSA_Certification_Authority.pem
Adding debian:vTrus_ECC_Root_CA.pem
Adding debian:vTrus_Root_CA.pem
Adding debian:XRamp_Global_CA_Root.pem
done.
Setting up maven (3.8.7-2) ...
update-alternatives: using /usr/share/maven/bin/mvn to provide /usr/bin/mvn (mvn) in auto mode
Setting up default-jre-headless (2:1.21-75+exp1) ...
victrespy@victrespy-ubuntu:~/Desktop/FP/Despliegue/Practica3/PracticaVictor/src/main$ mvn --version

Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.9, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "6.14.0-37-generic", arch: "amd64", family: "unix"
```

Paso 4

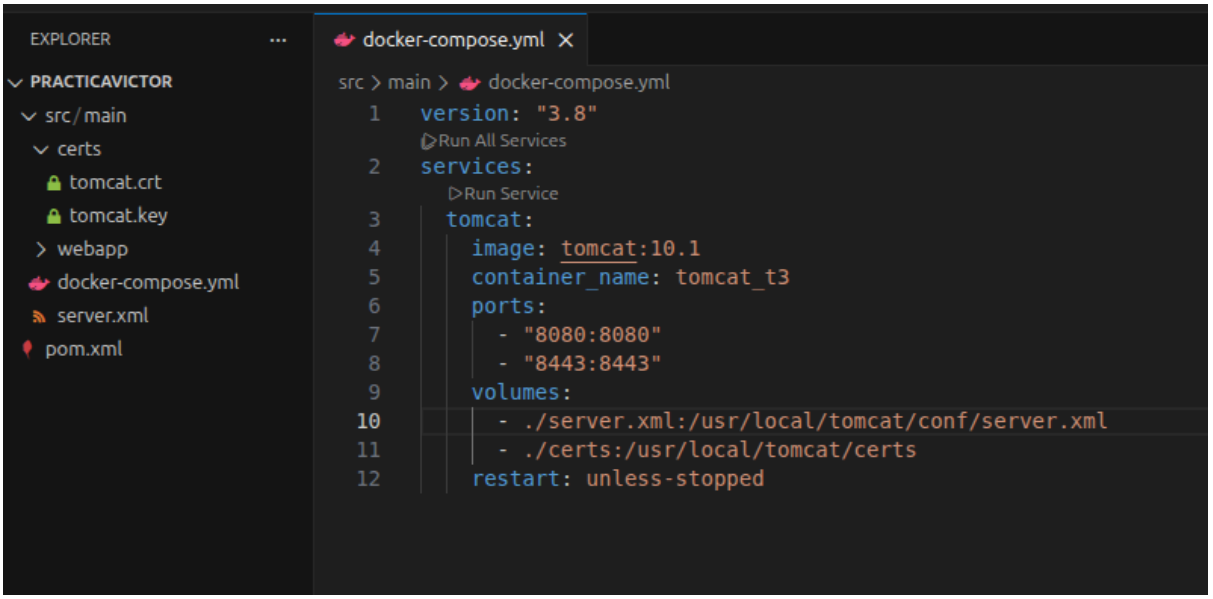
docker-compose.yml | pom.xml

pom.xml

```xml
<!--
 pom.xml = configuración de Maven.
 Maven se encarga de:
 - Descargar dependencias
 - Compilar el código Java
 - Empaquetar la aplicación como WAR para desplegar en Tomcat
-->
<project xmlns="http://maven.apache.org/POM/4.0.0">
 <modelVersion>4.0.0</modelVersion>

 <groupId>calculadora</groupId> <!-- Identidad del proyecto -->
 <artifactId>calculadora</artifactId> <!-- artifactId: nombre del proyecto/artefacto -->
 <version>1.0.0</version>
 <packaging>war</packaging> <!-- packaging: indica que el resultado final será un WAR -->

 <properties>
    <!--
    Versión de Java con la que compilamos.
    Recomendación docente: usar 17 para evitar diferencias entre equipos.
    -->
    <maven.compiler.release>17</maven.compiler.release>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
 </properties>


 <dependencies>
    <!--
    API de Servlets Jakarta (para Tomcat 10).
    scope=provided significa:
    - Se necesita para COMPILAR
    - NO se empaqueta dentro del WAR
    - Porque Tomcat ya incluye esta librería en el servidor.
    -->
    <dependency>
        <groupId>jakarta.servlet</groupId>
        <artifactId>jakarta.servlet-api</artifactId>
        <version>6.0.0</version>
        <scope>provided</scope>
```

Paso 5

Añado el server.xml y le pongo el volumen correspondiente a la receta de docker



Paso 6

```
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true"
           maxParameterCount="1000"
           >
    <UpgradeProtocol className="org.apache.coyote.http2.Http2Protocol" />
    <SSLHostConfig>
      <Certificate certificateFile="/usr/local/tomcat/certs/tomcat.crt"
                   certificateKeyFile="/usr/local/tomcat/certs/tomcat.key"
                   type="RSA" />
    </SSLHostConfig>
</Connector>
```

Paso 7

Creo los archivos de la calculadora y el index

```
EXPLORER
PRACTICAVICTOR
  src / main
    certs
    java
      CalculadoraServlet.java
    webapp
      index.jsp
  docker-compose.yml
  server.xml
  pom.xml

index.jsp
src > main > webapp > index.jsp > ? > ? > html > body > script > form.addEventListener("submit") callback
1  <%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
2  <%
3    //Evita que el navegador guarde la página en caché.
4    response.setHeader("Cache-Control","no-store");
5
6    //Comprobamos si hay un usuario autenticado.
7    if (request.getRemoteUser() == null) { response.sendRedirect("login.jsp"); r
8
9    //Comprobamos los roles del usuario
10   boolean basica = request.isUserInRole("calc_basic");
11   boolean full   = request.isUserInRole("calc_full");
12 %>
13 <!DOCTYPE html>
14 <html lang="es">
15 <head>
16   <meta charset="UTF-8">
17   <title>Calculadora JSP</title>
18   <link rel="stylesheet" href="css/estilo.css">
19 </head>
20 <body>
21
22 <div class="contenedor">
23   <!-- Cabecera con el titulo y el botón de cerrar sesión -->
24   <div class="cabecera">
25     <h2>Calculadora JSP</h2>
26     <!--Formulario para cerrar sesión.Se envia por POST a logout.jsp-->
27     <form method="post" action="<%=request.getContextPath()%>/logout.jsp">
28       <button type="submit">Cerrar sesión</button>
29     </form>
30   </div>
31   <!--Formulario principal de la calculadora.-->
32   <form id="formCalc" autocomplete="off">
33     <label>Número 1</label>

CalculadoraServlet.java
src > main > java > CalculadoraServlet.java
24   public class CalculadoraServlet extends HttpServlet {
34       protected void doPost(HttpServletRequest request, HttpServletResponse resp
68       }
69
70       //Comprobamos los roles del usuario:
71       boolean basica = request.isUserInRole("calc_basic");
72       boolean full   = request.isUserInRole("calc_full");
73
74       /*
75         Reglas de autorización:
76         - Los usuarios "full" pueden hacer cualquier operación
77         - Los usuarios "basic" SOLO pueden hacer "suma"
78       */
79       if (!(full || (basica && "suma".equals(op)))) {
80           out.print("{\"ok\":false,\"message\":\"Acceso denegado.\"}");
81           return;
82       }
83
84       //Realizamos la operación solicitada.
85       double r;
86       switch (op) {
87           case "suma": r = a + b; break;
88           case "resta": r = a - b; break;
89           default:
90               out.print("{\"ok\":false,\"message\":\"Operación no válida.\"}
91               return;
92       }
93
94       /*Devolvemos el resultado en formato JSON.
95       El frontend (JavaScript) leerá este mensaje y lo mostrará en pantalla.
96       out.print("{\"ok\":true,\"message\":\"Resultado: " + r + "\"}");
97   }
98 }
```

Paso 8

Me da un error al hacer el comando mvn clean package

```
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/collections/google-collections/1.0/google-collections-1.0.jar (640 kB at 1.3 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/3.8.2/junit-3.8.2.jar (121 kB at 248 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-logging/commons-logging-api/1.1/commons-logging-api-1.1.jar (45 kB at 85 kB/s)
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/target/classes
[INFO] -------------------------------------------------------------
[ERROR] COMPILATION ERROR :
[INFO] -------------------------------------------------------------
[ERROR] Source option 5 is no longer supported. Use 8 or later.
[ERROR] Target option 5 is no longer supported. Use 8 or later.
[INFO] 2 errors
[INFO] -------------------------------------------------------------
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  15.683 s
[INFO] Finished at: 2026-02-02T13:03:58+01:00
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.1:compile (default-compile) on project calculadora: Compilation failure: Compilation failure:
[ERROR] Source option 5 is no longer supported. Use 8 or later.
[ERROR] Target option 5 is no longer supported. Use 8 or later.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException
```

El fallo es por problema del pom.xml, le hago un cambio y vuelvo a ejecutar el comando. Habia que poner la version de java correcta, que no coincidia

Sige dando error

```
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ calculadora ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ calculadora ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug release 21] to target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.590 s
[INFO] Finished at: 2026-02-02T13:16:19+01:00
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.11.0:compile (default-compile) on project calculadora: Fatal error compiling: error: release version
pported -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoExecutionException
```

porque no reconoce la version de java que uso, el problema es que el sistema tiene instalado el JRE (entorno de ejecución) pero le falta el JDK (kit de desarrollo), que es el que contiene el compilador de Java

```
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-toolchain/2.0.9/maven-toolchain-2.0.9.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.8/plexus-utils-3.0.8.jar (232 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-annotations/3.1/maven-plugin-annotations-3.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/maven-surefire-common/2.12.4/maven-surefire-common-2.12.4.jar (263 kB at 1.2 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/reporting/maven-reporting-api/2.0.9/maven-reporting-api-2.0.9.jar (10 kB at 46 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.1/commons-lang3-3.1.jar (316 kB at 1.4 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-toolchain/2.0.9/maven-toolchain-2.0.9.jar (38 kB at 150 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-annotations/3.1/maven-plugin-annotations-3.1.jar (14 kB at 50 kB/s)
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (defa
Focus folder in explorer (ctrl + click)
[INFO] Packaging webapp
[INFO] Assembling webapp [calculadora] in [/home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/target/calculadora]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/src/main/webapp]
[INFO] Building war: /home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/target/calculadora.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.345 s
[INFO] Finished at: 2026-02-02T13:26:54+01:00
[INFO] ------------------------------------------------------------------------
```

ya funciona

actualizo el docker y accedo al navegador

```
ain > docker-compose.yml
version: "3.8"
Run All Services
services:
  Run Service
  tomcat:
    image: tomcat:10.1
    container_name: tomcat_t3
    ports:
      - "8080:8080"
      - "8443:8443"
    volumes:
      - ./server.xml:/usr/local/tomcat/conf/server.xml
      - ./certs:/usr/local/tomcat/certs
      - ../../target/calculadora.war:/usr/local/tomcat/webapps/calculadora.war
    restart: unless-stopped
```

Iniciar sesión

Usuario basic_user Contraseña ••• Entrar

Usuarios de prueba:

- **basic_user** / 123
- **full_user** / 321

Paso 9

docker-compose.yml | tomcat-users.xml | web.xml | login.jsp | logout.jsp | login-error.jsp | server.xml | pom.xml

src > main > tomcat-users.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<tomcat-users>

  <!-- Roles de la calculadora -->
  <role rolename="calc_basic"/>
  <role rolename="calc_full"/>

  <!-- Usuario con calculadora básica -->
  <user username="basic_user" password="123" roles="calc_basic"/>

  <!-- Usuario con calculadora completa -->
  <user username="full_user" password="321" roles="calc_full"/>

</tomcat-users>
```

docker-compose.yml | tomcat-users.xml | web.xml | login.jsp | logout.jsp | login-error.jsp | server.xml | pom.xml

src > main > docker-compose.yml

```yaml
version: "3.8"
Run All Services
services:
  Run Service
  tomcat:
    The name that will be given to the container
    container_name: tomcat_t3
    ports:
      - "8080:8080"
      - "8443:8443"
    volumes:
      - ./server.xml:/usr/local/tomcat/conf/server.xml
      - ./certs:/usr/local/tomcat/certs
      - ../../target/calculadora.war:/usr/local/tomcat/webapps/calculadora.war
      - ./tomcat-users.xml:/usr/local/tomcat/conf/tomcat-users.xml
    restart: unless-stopped
```

Me da un error 408
Cambio el action del form en el login

```html
<div class="contenedor">
  <h2>Iniciar sesión</h2>

  <form method="POST" action="j_security_check" autocomplete="off">
    <label>Usuario</label>
    <input name="j_username" required
     oninput="this.value=this.value.replace(/\s+/g,'')">

    <label>Contraseña</label>
    <input type="password" name="j_password" required
     oninput="this.value=this.value.replace(/\s+/g,'')">

    <button type="submit">Entrar</button>
  </form>

  <div class="nota">
    <p><b>Usuarios de prueba:</b></p>
```

tengo que volver a ejecutar el mvn clean pacage pero me da fallo

```
usua5pc15@A5PC15:~/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/src/main$ mvn clean package
[INFO] Scanning for projects...
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.070 s
[INFO] Finished at: 2026-02-02T13:53:15+01:00
[INFO] ------------------------------------------------------------------------
[ERROR] The goal you specified requires a project to execute but there is no POM in this directory (/home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/Practica
rify you invoked Maven from the correct directory. -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MissingProjectException
usua5pc15@A5PC15:~/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/src/main$
Ln 15, Col 1   S
```

pffffff… es por que no estoy en la carpeta correcta para ejecutar el comando, te juro que me estoy volviendo loco…

```
usua5pc15@A5PC15:~/Escritorio/Victor/Despliegue/Practica3/PracticaVictor$ mvn clean package
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ calculadora ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ calculadora ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ calculadora ---
[INFO] Packaging webapp
[INFO] Assembling webapp [calculadora] in [/home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/target/calculadora]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/src/main/webapp]
[INFO] Building war: /home/usua5pc15/Escritorio/Victor/Despliegue/Practica3/PracticaVictor/target/calculadora.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.576 s
[INFO] Finished at: 2026-02-02T13:55:33+01:00
[INFO] ------------------------------------------------------------------------
```

ahora si…

levanto el docker y ya si funciona

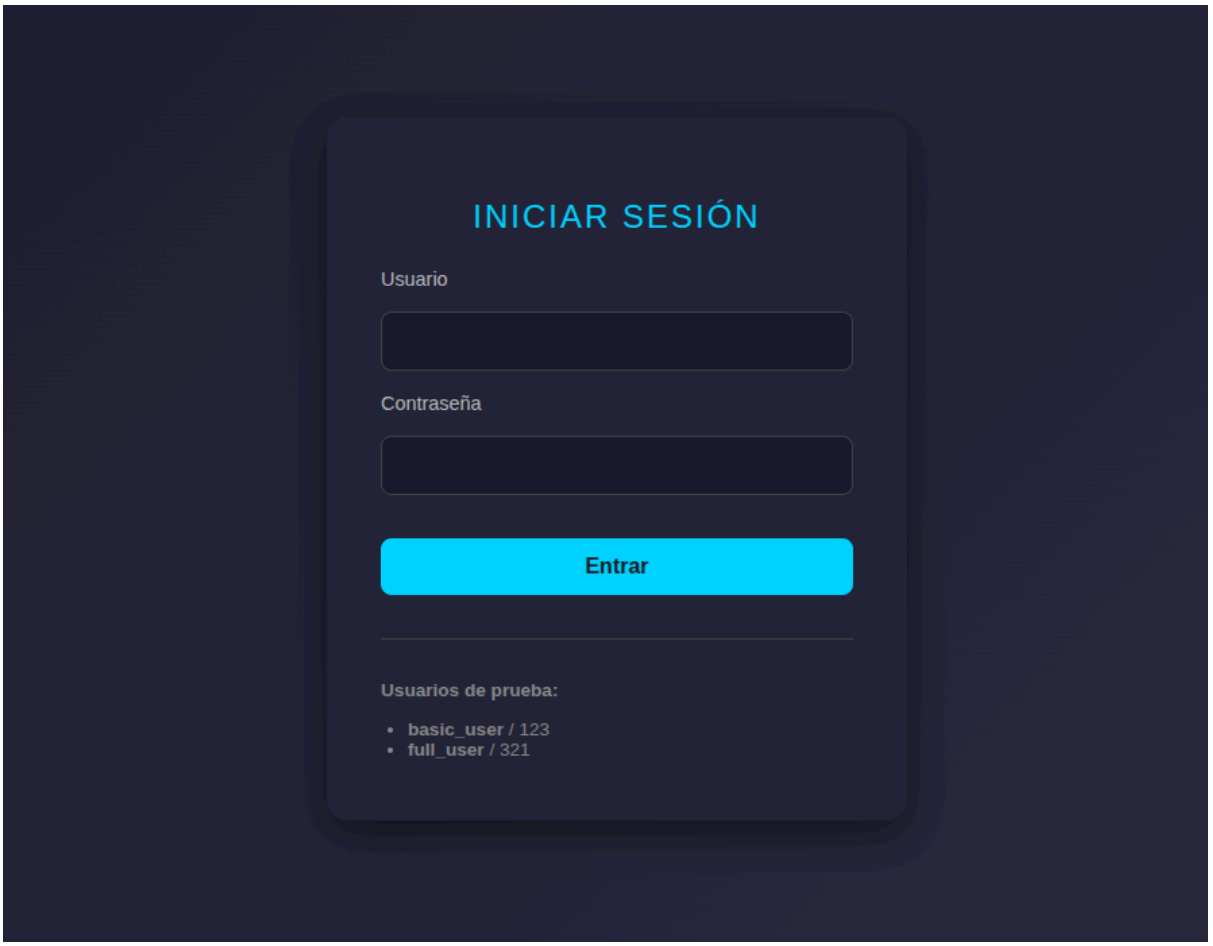
Calculadora JSP

Cerrar sesión

Número 1 [ ] Número 2 [ ] Operación [Suma] [Calcular]

Extras

Cambio los estilos



tambien añado nuevas operaciones para hacerlo mas completo